

PROJECT DEVELOPMENT REPORT

Vibe Coding & AWS Elastic Beanstalk Deployment | IBM & Bharat Cares

Live AWS Elastic Beanstalk URL: <http://StudyMate-AI-env.eba-6yuyfhhhu.ap-southeast-2.elasticbeanstalk.com>

1. Application Overview & Tech Stack

StudyMate.AI is a full-stack, AI-powered study platform built using modern Vibe Coding techniques, containerized with Docker, and deployed on AWS Elastic Beanstalk.

Tech Stack Breakdown:

- Frontend: React 19, Vite, Tailwind CSS v4, Motion animations, Lucide React icons.
- Backend: Node.js 20, Express.js framework, Esbuild bundler, Mammoth document parser.
- AI Model: Google Gemini 3.6 Flash (@google/genai SDK).
- Containerization: Docker (multi-stage Dockerfile based on Node.js 20 Alpine).
- Cloud Infrastructure: AWS Elastic Beanstalk (Docker on 64-bit Amazon Linux 2023), Single Instance Free Tier.

2. Prompting Strategy & Frameworks Used

Persona & System Prompting

The AI is assigned a pedagogical persona ('Expert Academic Tutor') instructed to maintain clarity, encouraging tone, and structured formatting.

Structured Output Schema (JSON)

For Quiz Generation, prompts enforce a rigid JSON schema enforcing question text, options array, correct option index, and detailed explanation to ensure flawless UI parsing.

3. Phase-by-Phase Development Summary

Phase 1: Conceptualization & UI Design

Designed a frosted-glass UI system with dark/light themes, responsive navigation, and tabbed workflow (Notes, Tutor, Quiz).

Phase 2: Backend & LLM Integration

Developed Express API endpoints (/api/simplify, /api/tutor, /api/quiz) ensuring Gemini API keys remain strictly secure on the server side.

Phase 3: Docker Containerization

Created a multi-stage Dockerfile packaging the full-stack app into a lightweight, isolated Docker container.

Phase 4: AWS Elastic Beanstalk Deployment

Launched AWS Elastic Beanstalk environment (studymate-ai-env) on Docker Platform, configuring environment variables (PORT, NODE_ENV, GEMINI_API_KEY).

4. Application Architecture

Client Browser % % (HTTP/Port 80)% % %° AWS Elastic Beanstalk Load Balancer/Nginx % % %° Docker Container Express App) % % (API Key / SDK)% % %° Google Gemini 3.6 API

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5. Challenges Encountered & Resolutions

Challenge 1: Dockerization & Elastic Beanstalk Memory Optimization

Initial container build compiled Vite and TypeScript on server during launch, leading to memory timeouts. Resolved by pre-building production dist bundle and using a lightweight Dockerfile copying dist directly.

Challenge 2: Environment Variable Security

Ensured API keys are never bundled in client code. Configured GEMINI_API_KEY as an AWS Elastic Beanstalk Environment Property.

6. Key Learnings & Reflection

This project demonstrated end-to-end Vibe Coding methodology—architecting, containerizing with Docker, and deploying a functional AI web application on AWS Elastic Beanstalk Free Tier with zero cost and maximum security.